CONVERSION, OBVERSION, and CONTRAPOSITION

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What is an inference? Differentiate mediate from immediate inference.

Inference

- comes from the latin word “in-ferre” which basically means “to bear” or “to produce”. This etymology suggests that the mental process of inference involves the production of a new proposition directly related with the preceding propositions. The truth-value of this new emerging proposition implies the truth of the previous propositions.
- Two types of inference:
  - Mediate inference
  - Immediate inference

Distinction Between ...

Mediate inference	Immediate inference
There are 3 or more propositions.	There are only 2 propositions.
There is a middle proposition.	There is no middle proposition.
The third proposition is the conclusion.	The second proposition is the equivalent proposition.
The conclusion has new meaning or truth	The equivalent proposition has the same truth or meaning as the given.

Examples:

Mediate inference	Immediate inference
•All Filipinos are Asians. •All Davaoeños are Filipinos. •Therefore, all Davaoeños are Asians.	•Original proposition: Some students are scholars. •Equivalent proposition: Some scholars are students.

Enumerate the three rules of obversion and explain each.

Obversion

- comes from Latin "ob" meaning "before" or "toward" and "vertere" which means "to turn".
- is a process of reduction involving two changes. These changes occur in the quality of proposition and in the status of the predicate term.

3 Rules of Obversion:

1. Retain the subject and the quantity of the proposition.
   - A universal proposition remains a universal proposition and a particular proposition remains a particular proposition.
2. Change the quality of the proposition.
   - An Affirmative proposition will be changed into a negative proposition , and a negative proposition into an affirmative proposition.
3. Substitute the predicate by its contradictory or complementary term.
   - There are some words in English language which have a corresponding contradictory term, i.e., invisible - visible, ineligible – eligible, illegible – legible.
   - Complementary term is a contradictory term that uses the appropriate prefix "non", for example "black" is "non-black".

Obversion: Changing the quality of a proposition and then replacing the predicate with its complement.

Step 1: Change the quality: We learned how to change the quality of a proposition. Remember, **quality** refers to whether the proposition is "affirmative" or "negative." For instance, changing the quality of "All S are P" would give us "No S are P."

Step 2: Replace predicate with its complement: The **complement** of a predicate class is the set of everything that is NOT a member of the predicate class. So, for instance, if our predicate class is "dogs", then the COMPLEMENT class includes everything from cats to chairs to clouds to stars—in short, everything that is not a dog (all "non-dogs").

Example: To illustrate, let's obvert "All dogs are mammals."

Step 1: First, change the quality. The proposition becomes "No dogs are mammals."

Step 2: Next, replace the predicate with its complement. The complement of "mammals" is the set of all things that are NOT mammals—in other words, "nonmammals"—so the proposition becomes, "No dogs are non-mammals."

"All dogs are mammals" tells us that ALL of the dogs are INSIDE of the mammal class. But, "No dogs are non-mammals" tells us that NONE of the dogs are OUTSIDE of the mammal class. These are both saying the same thing. So, it appears that the obverted proposition has preserved the truth value of the original proposition.

As it turns out, obversion ALWAYS preserves the truth value for all four kinds of categorical proposition. Below is an illustration of obversion for each of these four kinds, with specific examples:

What do we mean by "do not extend any term"?

"Do Not Extend Any Term"

- Is one of the rules under Conversion.
- Quantity of the terms in the proposition must not be affected in the process of conversion. Meaning, terms shouldn't extend or increase in their quantity or extension of objects. If a term in a convertend is universal, it should still be universal in converse; and if the term is particular in convertend, it must still be particular in converse.
- There are instances where universal terms in convertend becomes particular in converse is allowed, we call this as "conversion by limitation" or "partial conversion".

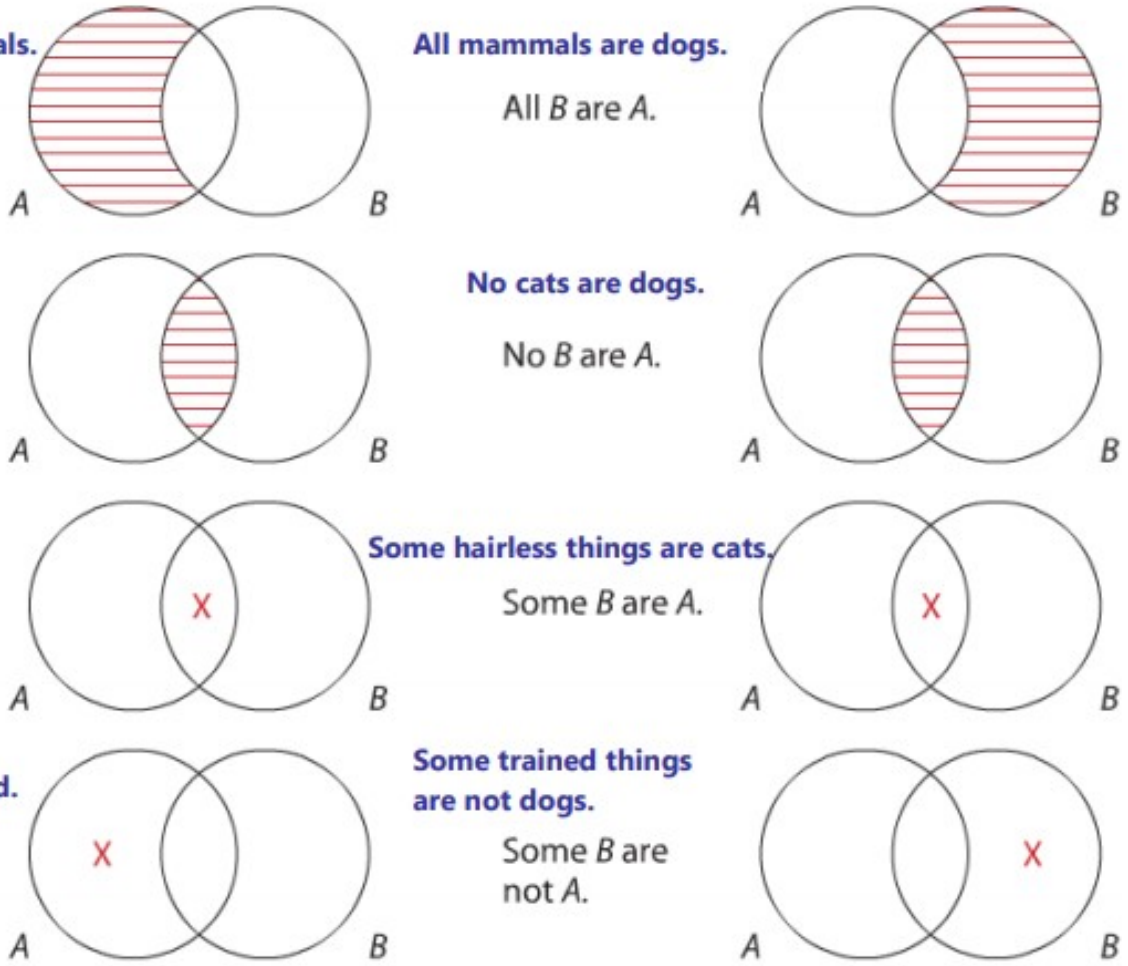
Conversion: Swapping the subject with the predicate.

For instance, consider some cases of conversion:

- "No **birds** are **bumblebees**." **CONVERSION→** "No **bumblebees** are **birds**."
- "Some **birds** are **flying creatures**." **CONVERSION→** "Some **flying creatures** are **birds**."

Below is a chart depicting how each of the four types of categorical proposition are converted. The **statements about dogs and cats** provide specific examples:

Given statement form		Converse	
All dogs are mammals.	All *A* are *B*.	**All mammals are dogs.**	All *B* are *A*.
No dogs are cats.	No *A* are *B*.	**No cats are dogs.**	No *B* are *A*.
Some cats are hairless.	Some *A* are *B*.	**Some hairless things are cats.**	Some *B* are *A*.
Some dogs are not trained.	Some *A* are not *B*.	**Some trained things are not dogs.**	Some *B* are not *A*.



Note: (1) The conversion of the middle two examples above ***yields a Venn diagram that is exactly the same****. (2) The conversion of the top and the bottom examples* ***yields a Venn diagram that is NOT the same****.*

There are four types of categorical proposition, each of which is given a vowel letter A, **E**, I and O.

A way of remembering these is:

Affirmative universal,
nE**g**ative universal,
affI**r**mative particular and
nO**g**ative particular.

Is it legitimate to convert O propositions? Explain your answer.

Conversion of O Proposition

- Basically, conversion does not hold for O propositions because it violates the rule on non-extension of terms and changes the truth in its entirety.

  Example:

  Some males are not fathers.

  Converted: Some fathers are not males.

- The convertend is true but the converse is false.
- The quantity of the term "males" had also increased because the subject term "males" is particular but it became universal in the predicate term.

How would you contrast simple from partial conversion?

Simple Conversion and Partial Conversion

- Simple and partial conversion may vary in few things, simple conversion retains the entirety if the truth-value, while partial conversion need to reduce one of the terms by applying 2 rules as remedy in order not to violate rules of conversion.
- However, the very important thing to remember is that we should be cautious in conversion, the truth or falsity of the convertend should still be retained in the converse.

There are four types of categorical proposition, each of which is given a vowel letter A, **E**, I and O.

A way of remembering these is:

Affirmative universal,
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Simple Conversion

- Only E and I propositions are legitimate for simple or absolute conversions.

Examples for Simple Conversions:

- E Proposition
  - No roses are tulips. → No tulips are roses.
  - No apes are animals. → No animals are apes.

- I Proposition
  - Some snakes are venomous creatures. → Some venomous creatures are snakes.
  - Some jewels are gems. → Some gems are jewels.

Partial Conversion

- A propositions can sometimes be converted under partial conversions. Partial conversion does not totally retain the truth or falsity of the proposition once converted because it reduces one of the terms. To avoid changing the truth-value into false in converse, there are 2 rules needed to be applied:

1. Sub-altern the given convertend. The sub-altern is the A and the sub-alternate is the I.
2. Convert the product sub-alternate I to converse I.

Example:

All iron are minerals.

→Simple conversion: All minerals are iron. ✘

→Apply rule 1 : Some iron are minerals.

→Apply rule 2 : Some minerals are iron. ✔

What are the types of conversion applicable to A propositions with predicate terms as definitions?

A Propositions With Predicate Terms As Definitions

- A proposition with predicate terms as definitions may be solely converted by simple conversion (A to A) or by partial conversion (A to I).

  Example:

  All mothers are women who gave birth.

→Simple conversion: All women who gave birth are mothers. ✔

→Partial conversion: Some women who gave birth are mothers. ✔

Conversion of (E) and (I) preserves truth value: We say that the converted proposition of the middle two examples above **"preserves the truth value"** of the original proposition.

This happens whenever the conversion of a proposition yields a Venn diagram that is exactly the same as the converted proposition.

In other words, if the original proposition is TRUE, then the conversion of it will ALSO be true (and, similarly, if the original proposition is false, then the conversion of it will also be false).

Example #1: Since it is TRUE that "No **dogs** are **cats**" then it is automatically ALSO true that "No **cats** are **dogs**."

Example #2: Since it is FALSE that "No **dogs** are **mammals**" then it is automatically ALSO false that "No **mammals** are **dogs**."

There is any conversion for false A propositions. Why?

Conversion For False A Propositions

- There is no conversion that can be applied for false A proposition whether it be of simple or partial conversion because it affects and change the truth-value of the convertend.
- As stated in the one of the rules of sub-alternation; if the universal is false, the particular is undetermined or doubtful.

  Example:

  All mangoes are apples.

  - Sub-alternate: Some mangoes are apples. → Undetermined or doubtful.

Conversion of (A) and (O) does not preserve truth value: But, if the conversion of a proposition does NOT yield a Venn diagram that is exactly the same as the converted proposition, then we cannot know for sure whether or not the converted proposition has the same truth value as the original proposition.

Example #1: If "All **dogs** are **mammals**" is true, it is NOT automatically true that "All **mammals** are **dogs**."

Example #2: Similarly, if "Some **dogs** are not **trained creatures**" is true, it is NOT automatically true that "Some **trained creatures** are not **dogs**."

If we DID claim that the converted proposition preserved the truth value of the original proposition in these two cases, we would be making a mistake known as **"the fallacy of illicit conversion."**

Here is a chart that summarizes what we have just said about conversion:

Proposition	Converse	Preserves Truth Value?
(A) All S are P.	All P are S.	No
(E) No S are P.	No P are S.	Yes
(I) Some S are P.	Some P are S.	Yes
(O) Some S are not P.	Some P are not S.	No

Illustrate with examples the rechecking pattern under contraposition.

Contraposition

- The process of suction which combines conversion and obversion.
- Formed by 2 steps:
  1. Switching the subject term and predicate term as in conversion.
  2. Substituting both subject term and the predicate terms with contradictory or complementary terms as in obversion. The original proposition is the contraponend while the equivalent proposition is the contrapositive.
- It is legitimate for A and O propositions (simple) and E (partial) but not for I.

3 Operations We Must Apply Upon Checking Legitimacy

1.) Obvert the original proposition.

2.) Convert the resulting proposition in step 1.

3.) Obvert the resulting proposition in step 2.

Note:

The resulting proposition in step 3 is the contrapositive.

A propositions – Simple Contrapositions

- Example:

All men are mortal beings.

Apply 2 steps of contraposition:

→All immortal beings are non-men. ✔

To re-check the resulting contrapositive, apply the 3 steps:

- Obverse :No men are immortal beings.
- Converse:No immortal beings are men.
- Obverse:All immortal beings are non-men.(Contrapositive) ✔

O Proposition – Simple Contrapositions

- Example:

Contraponend: Some birds are flying creatures.

Contrapositive: Some non-flying creatures are not non-birds.

- Re-checking order:

Obverse: Some birds are non-flying creatures.

Converse: Some non-flying creatures are birds.

Obverse: Some non-flying creatures are not non-birds.(Contrapositive) ✔

E proposition – Partial contrapositions

- Example:

No spirits are mortal beings.

Contraposition → Some immortal beings are not non-spirits. ✔

- Re-checking:

Contraponend → No spirits are mortal beings.

Obverse: All spirits are immortal beings.

Converse: Some immortal beings are spirits.

Obverse: Some immortal beings are not non-spirits. (Contrapositive) ✔

Contraposition: Last but not least, the third sort of swap.

Contraposition: Performing an conversion on a proposition (i.e., swapping the subject with the predicate) and then replacing both the subject and the predicate terms with their complements.

Example: Let's try one: "All dogs are mammals."

Step 1: Obversion: First, we obvert it. That is, we replace the subject and the predicate to get, "**All mammals are dogs**."

Step 2: Replace subject and predicate with complements: Next, we replace both terms (subject and predicate) with their complements. So, "mammals" becomes "non-mammals", while "dogs" becomes "non-dogs." The end result is this: "**All non-mammals are non-dogs**."

There is no contraposition for I propositions. Why? Elaborate your answer.

Contraposition For I Propositions

- There is no contraposition for I propositions.
- The procedure is unavailable for I propositions because it can disrupt the order of re-checking.

Example:

- Contraponend: Some computers areY2K compliant.
- Contrapositive: Some non-Y2K compliant are non-computers.

Re-checking:

- Obverse: Some computers are non-Y2K compliant.
- Converse: (No converse for O propositions).

(A)and (O) preserve truth value. (E) and (I) do not: So, this time, the top and bottom (A & O) preserve the truth value, while the middle two (E & I) do not.

As with obversion, a valid inference CAN immediately be derived for two of the four types of categorical proposition, but NOT for the other two. For instance, if you have an (A) or an (O) proposition as a true premise, you can contrapose it and automatically conclude that it is also true. On the other hand, if you have a true (E) or an (I) proposition, and you contrapose it, and then claim that contraposed proposition is also true, you are making a mistake known as **"the fallacy of illicit contraposition."**

Here is a chart that summarizes what we have just said about contraposition:

Proposition	Contraposition	Preserves Truth Value?
(A) All S are P.	All non-P are non-S.	Yes
(E) No S are P.	No non-P are non-S.	No
(I) Some S are P.	Some non-P are non-S.	No
(O) Some S are not P.	Some non-P are not non-S.	Yes

SUMMARY

There are four types of categorical proposition, each of which is given a vowel letter A, **E**, I and O.

A way of remembering these is:

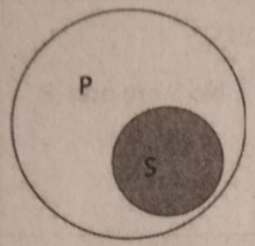
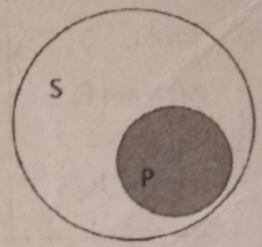
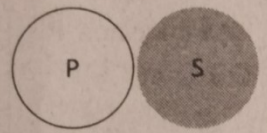
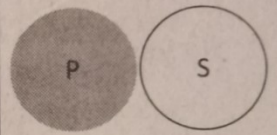
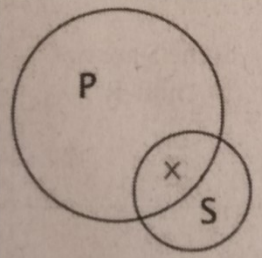
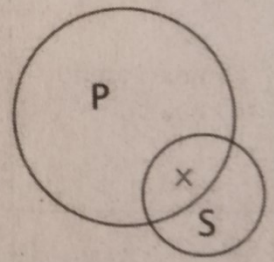
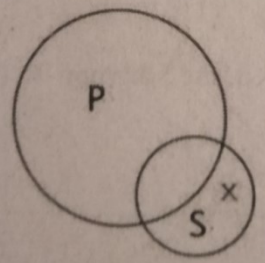
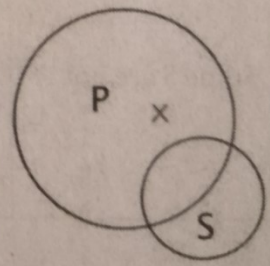
Affirmative universal,
nE**g**ative universal,
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nO**g**ative particular.

Conversion is the inference in which the subject and predicate are interchanged. In modern logic it is only valid for the **E** and **I** propositions. The valid converse is logically equivalent to the original proposition. In traditional logic, the **A** proposition has a converse by limitation which is the subaltern of the invalid **A**-converse; i.e., the corresponding **I** proposition. The converse by limitation is implied by the original but is not (usually) equivalent to it.

ORIGINAL	CONVERSE	VALID?	BY LIMITATION
All S are P	All P are S	no	Some P are S
No S are P	No P are S	yes	
Some S are P	Some P are S	yes	
Some S are not P	Some P are not S	no	

Solution:

Converse

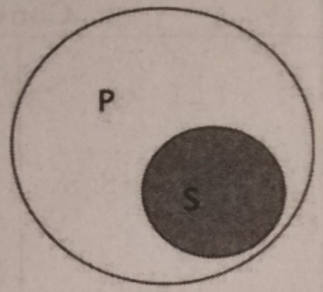
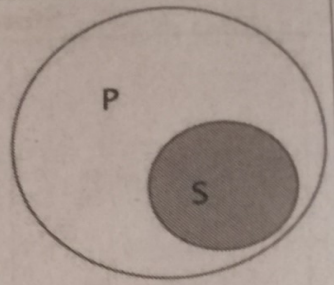
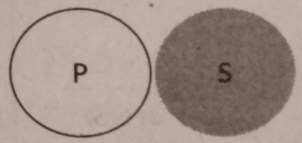
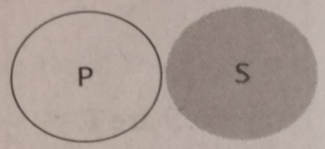
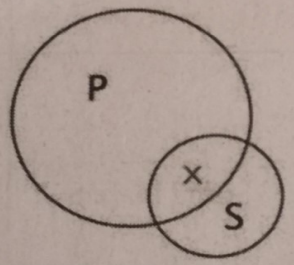
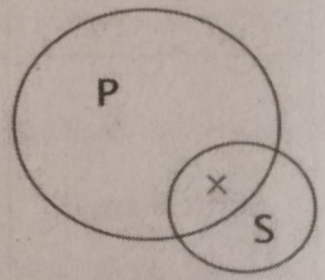
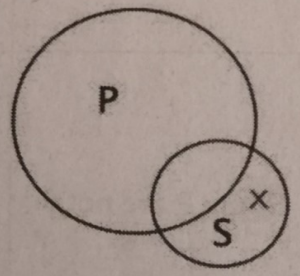
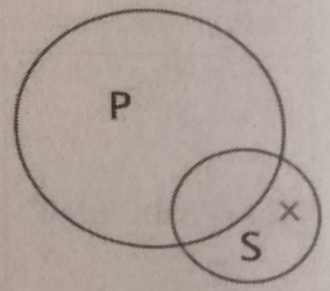
Statement		Converse	
All S are P.		All P are S.	
No S are P.		No P are S.	
Some S are P.		Some P are S.	
Some S are not P.		Some P are not S.	

Based on the corresponding Euler diagrams, the converse of an E-proposition and I-proposition preserves the truth value. This means that whenever E-proposition or I-proposition is true (or false), the conversion is also true (or false).

Obversion is the inference in which the quality of the proposition is changed and the predicate is interchanged with its complement. It is valid for all four forms. The obverse is logically equivalent to the original proposition.

ORIGINAL	OBVERSE	VALID?
All S are P	No S are nonP	yes
No S are P	All S are nonP	yes
Some S are P	Some S are not nonP	yes
Some S are not P	Some S are nonP	yes

Obversion

Statement		Obverse	
All S are P.		No S are non-P.	
No S are P.		All S are non-P.	
Some S are P.		Some S are not non-P.	
Some S are not P.		Some S are non-P.	

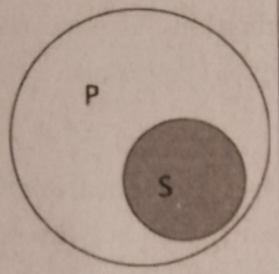
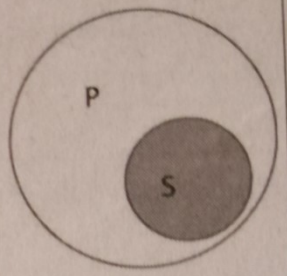
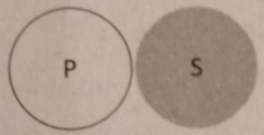
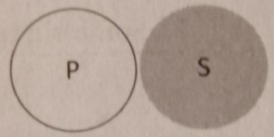
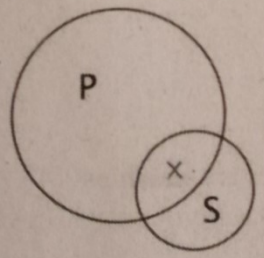
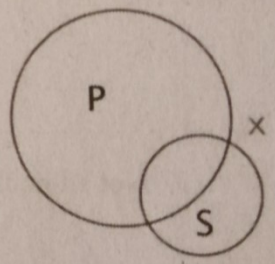
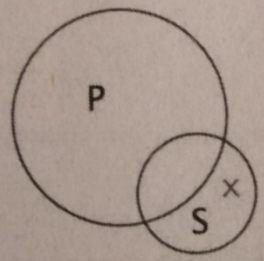
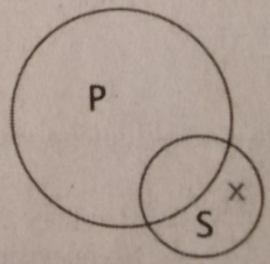
Notice that the obversion of the four types of categorical syllogisms yields the same Euler diagrams; hence, the truth value is preserved.

Contraposition is the inference in which the subject is interchanged with the complement of the predicate and *the predicate is interchanged with the complement of the subject*. In modern logic it is only valid for the **A** and **O** propositions. The valid contrapositive is logically equivalent to the original proposition. In traditional logic, the **E** proposition has a contrapositive by limitation which is the subaltern of the invalid **E**-contrapositive; i.e., the corresponding **O** proposition. The contrapositive by limitation is implied by the original but is not (usually) equivalent to it.

ORIGINAL	CONTRAPOSITIVE	VALID?	BY LIMITATION
All S are P	All nonP are nonS	yes	
No S are P	No nonP are nonS	no	Some nonP are not nonS
Some S are P	Some nonP are nonS	no	
Some S are not P	Some nonP are not nonS	yes	

Contraposition

The truth value of the A, E, and O-propositions are preserved in the contraposition. The Euler diagrams are shown in the table.

Statement		Contrapositive	
All S are P.		No non-P are S.	
No S are P.		All non-P are S.	
Some S are P.		Some non-P are not S.	
Some S are not P.		Some non-P are S.	

ACITIVITY:

C. **Directions**: Derive another proposition with the same meaning but of different quality using conversion, obversion, and contraposition.

1. All juvenile delinquents are maladjusted individuals.

2. Some diamonds are precious stones.

3. Some politicians are not philosophers.

4. No medicines that can be purchased without a doctor's prescription are habit-forming drugs.